

HIMALAYAN WHEELS

Full Stack Car Rental System

A Project Report

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September 2026

Acknowledgement

I acknowledge the maintainers of React, Node.js, Express, SQLite, Vite, and the other open-source tools used in Himalayan Wheels. Their software and public documentation provided the foundation for implementing and understanding this application. I also acknowledge the security and database reference materials cited in this report, which informed the design and its limitations.

This project brings together interface design, server development, database modeling, and verification in one working educational system. It is presented as a practical demonstration of a car rental booking workflow.

Abstract

Himalayan Wheels is a full-stack car rental demonstration designed for browsing vehicles and managing reservations in a Nepal-focused setting. The system presents a six-vehicle sample fleet, sample daily rates in Nepalese rupees, and pickup choices in Kathmandu, Pokhara, and Chitwan. Visitors can explore vehicle categories, search models, sort prices, and check availability for selected dates. Registered users can reserve a vehicle, review their own booking history, and cancel a reservation before its pickup date. Rental owners can sign in to review customer booking orders.

The frontend uses React and responsive CSS. An Express API implements authentication, booking rules, and authorization, while SQLite stores users, sessions, cars, and bookings. The server calculates the rental total from its own daily rate and validates the date range. A transaction encloses the final availability check and insertion so that overlapping confirmed bookings for the same vehicle are rejected.

Verification covered account isolation, authoritative pricing, conflicting reservations, adjacent date ranges, cancellation, persistence after restart, and browser interactions on desktop and mobile. The resulting application demonstrates an end-to-end reservation workflow. It does not process payments, verify driver documents, or operate a real rental fleet. Source code, setup instructions, test automation, screenshots, and this report are organized for publication in the project GitHub repository.

Keywords: React, car rental, reservation, Express, SQLite, date availability, full-stack web application.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CI	Continuous Integration
CSS	Cascading Style Sheets
DB	Database
EV	Electric Vehicle
FK	Foreign Key
HTTP / HTTPS	Hypertext Transfer Protocol / HTTP over TLS
JSON	JavaScript Object Notation
NPR	Nepalese Rupee
PK	Primary Key
QR	Quick Response
REST	Representational State Transfer
SQL	Structured Query Language
SVG	Scalable Vector Graphics
UI	User Interface
UUID	Universally Unique Identifier

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GitHub Repository Link and QR Code

The project repository is available at <https://github.com/Anish00079/Himalayan-Wheels->. The QR code below encodes this HTTPS address, which is suitable for opening in a browser. The SSH clone address supplied for the project is `git@github.com:Anish00079/Himalayan-Wheels-.git`.



Figure 1 Scan to open the Himalayan Wheels GitHub repository

The repository groups the React frontend, Express backend, database initializer, test suite, setup guide, deployment starting files, and academic report. The website preview at <https://anish00079.github.io/Himalayan-Wheels-/> runs on GitHub Pages. It uses a clearly labelled browser-only demo workspace without passwords or shared bookings. The full-stack build still requires an application host, HTTPS, and persistent database storage.

Introduction

A car rental workflow connects a travel plan to a particular vehicle over a defined period. A useful web interface must help the customer compare options, understand the price, and determine whether the selected vehicle can actually be reserved. Behind that interface, the system must distinguish one customer from another, retain bookings, and prevent conflicting allocations.

Himalayan Wheels models this workflow with a small Nepal-oriented sample fleet. It focuses on the transition from public vehicle discovery to authenticated reservation management. The client provides immediate interaction and price previews; the server remains responsible for deciding whether a booking is valid. This division makes the project suitable for demonstrating the relationship between presentation, business rules, and persistent data.

The intended users are visitors browsing the fleet, registered customers managing their own bookings, and rental owners reviewing customer orders. The owner dashboard is read-only and uses backend role checks. It is not a commercial fleet-management system.

Problem Statement

A booking interface is incomplete if it only displays cars and accepts form submissions. It must also resolve practical questions: Is the selected vehicle already reserved? Are the dates valid? Is the quoted price calculated from the correct rate? Can one customer access or cancel another customer's reservation? Does a confirmed booking remain available after a server restart?

When availability checks are separated from booking creation, two requests can both observe an available vehicle and produce conflicting reservations. When totals are accepted from the browser, the submitted price can be manipulated. When booking queries are not restricted by user identity, personal reservation information can be exposed. These are specific implementation problems addressed by this project; no claim is made about their prevalence in any particular rental business.

Objectives

The primary objective is to develop a working React-based car rental application with an authenticated backend and a persistent relational database. The implementation should support a complete customer journey and make its booking rules visible and testable.

- Provide a public fleet with model search, category selection, price sorting, and readable vehicle details.
- Check availability for valid date ranges of one to thirty days using the Nepal calendar.
- Register and authenticate users, maintain sessions, and restrict booking history and cancellation to the owner.
- Calculate each total on the server and store the daily rate, day count, and total as a booking snapshot.
- Prevent overlapping confirmed bookings through a transactional check and insertion.
- Allow cancellation before the pickup date and release availability while preserving the booking record.
- Verify key API rules, persistence, browser workflows, and mobile layout; provide a reproducible project report.

Acceptance criteria

The core workflow is accepted when a visitor can find a vehicle, create an account, confirm a valid reservation, reload the application, view the saved booking, and cancel it before pickup. Negative checks must show that invalid dates, price tampering, conflicting bookings, and unauthorized cancellation do not succeed.

Literature Review

This section reviews technical documentation relevant to the implemented system. It is a focused engineering review, rather than a systematic survey of academic research or a market comparison of rental providers.

Component based interfaces

React describes interfaces through components, event handlers, state, and conditional rendering [1]. These concepts support reusable vehicle cards, authentication forms, and reservation dialogs. Himalayan Wheels applies them to keep the selected trip, visible fleet, signed-in user, and booking screens synchronized. The benefit is a consistent interactive workflow without a full document reload after every action.

Server responsibilities and security

Express documentation explains application setup and production security practices, including transport security, input handling, and defensive response headers [2, 3]. This project places authorization and rental validation in the server instead of relying on hidden frontend controls. OWASP recommends salted, deliberately expensive password hashing [4]. The implementation uses scrypt with explicit work parameters, and stores session-token digests separately from password hashes. These controls are a foundation, not evidence of a complete security audit.

Relational persistence and reservation intervals

SQLite is an embedded relational database suited to local and modest single-server applications [5]. Its transaction documentation explains how an immediate transaction obtains a write transaction before dependent changes proceed [6]. Himalayan Wheels uses that mechanism around the availability check and booking insert. Node exposes the database through its built-in SQLite module [7], reducing the number of separately installed services needed for a classroom demonstration.

PostgreSQL documents range values and exclusion constraints as tools for expressing interval restrictions [8]. That approach provides a useful future direction if this project moves to a larger database. The current implementation instead expresses overlap as two comparisons over ISO date strings and applies that rule in one SQLite transaction.

Design conclusion

The review supports a three-part design: a component-based client, an authoritative API, and a relational store. The project prioritizes transparent booking rules and a small setup footprint. The resulting design is deliberately narrower than a commercial rental platform, which would also need operational inventory, payments, document checks, and customer support workflows.

Scope and Limitations

The implemented scope covers public discovery and private reservation management. Six seeded records represent six physical demo vehicles. Model specifications and daily rates are illustrative data, not verified manufacturer specifications or market quotations. Vehicle photos from Meromoto are included as local assets, with source details in Image-Credits.md. The website uses a grey, black, and white interface.

The customer selects one of three pickup cities and returns the vehicle to the same city. The application does not model where a vehicle is physically located, repositioning between cities, pickup times, driver services, maintenance blocks, or turnaround buffers. A return date is exclusive, so another rental can start on that date. These assumptions simplify the demonstration and must be revisited before commercial use.

The optional GitHub Pages preview uses local browser storage and a demo identity; it does not provide the authentication or shared-database guarantees of the full-stack application. No money, driving license, identity document, or insurance information is collected. Payment is described as due at pickup, but no actual rental service is issued. A read-only owner dashboard is included. Email verification, password reset, notifications, and broader staff operations remain outside the scope. SQLite is used on a single application server; horizontal scaling is outside the current scope.

Requirement Analysis

Table 1 Functional requirements

ID	Requirement	Acceptance evidence
FR1	Browse and compare the fleet	Six cars load; categories, search and price sorting work.
FR2	Check valid date availability	Unavailable cars are marked; invalid ranges are rejected.
FR3	Register and authenticate	Valid accounts receive a session; invalid credentials fail.
FR4	Create an owned reservation	Signed-in users receive a stored booking reference.
FR5	Calculate authoritative totals	Client-supplied totals are ignored; daily rate $\times$ days is stored.
FR6	Reject overlapping reservations	A second conflicting booking returns HTTP 409.
FR7	Manage private bookings	History and cancellation are limited to the booking owner.
FR8	Cancel future reservations	Cancellation changes status and releases availability.
FR9	Owner order dashboard	Only owners can read all customer booking orders.

Requirement Analysis and Technology Stack

Nonfunctional requirements

- Usability: labelled forms, visible errors, a responsive layout, keyboard-operable controls, and dialogs that restore focus.
- Integrity: transactional booking creation, foreign keys, validated date ranges, and server-owned pricing.
- Privacy: private booking queries, hashed passwords, hashed session tokens, HttpOnly cookies, and session expiry.
- Maintainability: separate client and server directories, reproducible dependencies, tests, readable formatting, and setup documentation.
- Portability: Node.js 24 or newer and a modern browser; no separate database service is required for local use.

Technology Stack

Table 2 Technology stack

Layer	Technology	Role
User interface	React 19.3.0	Component rendering, state, events and dialogs
Build tools	Vite 7.3.6	Development server and frontend production bundle
Server runtime	Node.js 24+	JavaScript execution, cryptography and SQLite API
API framework	Express 5.2.1	Routes, middleware, JSON responses and static serving
Database	SQLite through node:sqlite	Durable users, sessions, vehicle and booking records
Security middleware	Helmet 8.3.0; express-rate-limit 8.7.0	Response headers and authentication attempt limits
Visual interface	CSS, SVG; Lucide React 0.468.0	Responsive styling, vehicle photos and icons
Verification	Node test runner; Playwright browser checks	API assertions and end-to-end interaction checks
Source management	Git and GitHub Actions	Version history and automated test/build workflow

Exact package resolutions are recorded in package-lock.json. Vite provides a development server and a production build pipeline [9]. During development its proxy forwards API requests to Express; in the production build Express serves the generated frontend from the same origin as the API.

Methodology

Development followed an iterative implementation and verification process. Each iteration tied a visible customer action to a server rule and a stored record. The sequence began with the domain model and reservation rules, continued through the API and interface, and ended with negative tests, browser checks, and documentation.

- Define the journey: browse, select dates, sign in, review the price, confirm, view, and cancel.
- Model the data: separate users, sessions, cars, and bookings; define ownership and foreign keys.
- Implement server rules: validate dates, derive prices, protect endpoints, and control overlap in a transaction.
- Build the interface: responsive navigation, vehicle photo cards, dialogs, availability indicators, and booking history.
- Verify behavior: test successful and rejected requests, restart persistence, race attempts, and browser interaction.
- Package the result: retain source, dependency lockfile, setup instructions, screenshots, report, and CI workflow.

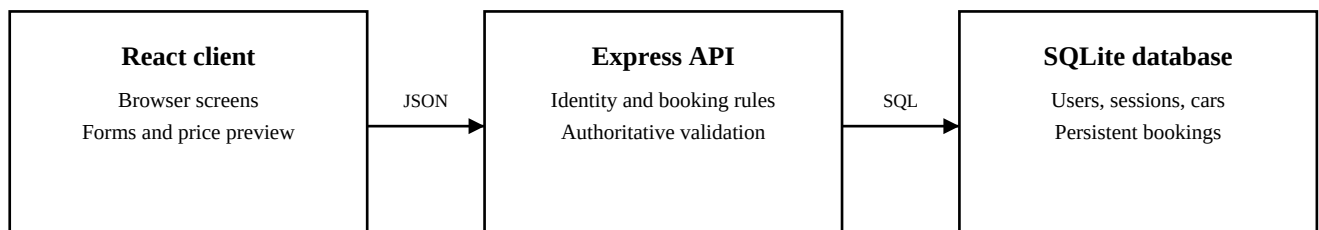


Figure 2 Same-origin application architecture

Request and data flow

The browser sends JSON requests to the API and includes the session cookie automatically. Public requests can retrieve the fleet. A protected route first resolves the session to a user. Booking creation then checks the car, city, dates, and conflicting confirmed bookings before committing the record. The response updates the client view and the booking history.

The application uses a same-origin deployment model. No cross-origin API access is enabled. Each state-changing browser request includes a custom verification header; the session cookie is HttpOnly and SameSite Strict. The production setting additionally requires secure cookies over HTTPS. These controls are implemented alongside server-side authorization, rather than replacing it.

Implementation of the Data Layer

The database initializer creates four related tables, enables foreign keys and write-ahead logging, and seeds the six demonstration vehicles idempotently. Existing accounts and reservations remain untouched when the server restarts. All runtime queries use bound parameters instead of concatenating user input into SQL.

Table 3 Database entities

Entity	Key fields	Relationships and purpose
users	id PK; name; email UNIQUE; password; role	Customer or owner role; stores a salted password hash.
sessions	token PK; user_id FK; expires	Maps a token digest to one user until the expiry time.
cars	id PK; name; category; seats; transmission; fuel; price; color; description	One record represents one reservable physical demo vehicle.
bookings	id PK; user_id FK; car_id FK; pickup; start_date; end_date; days; daily_rate; total; status; created_at	Stores ownership, interval, rate snapshot, total and cancellation status.

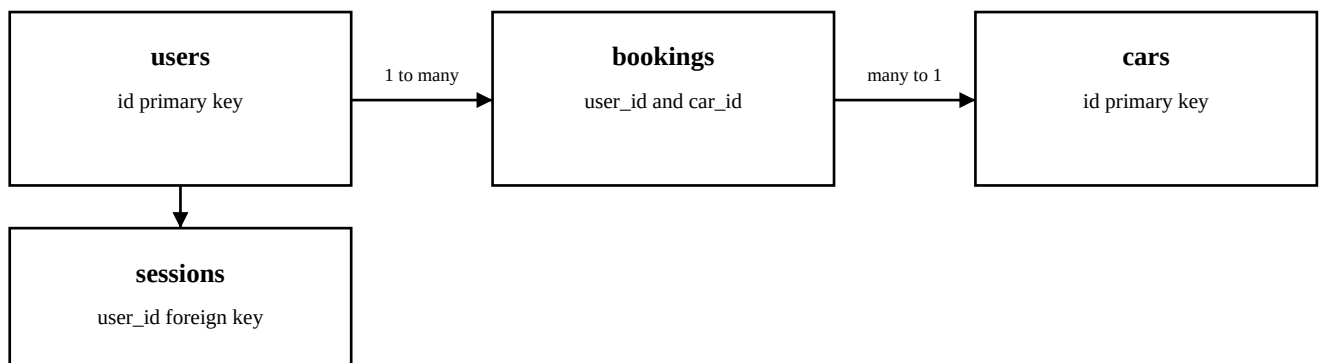


Figure 3 One user owns many bookings and sessions; one car has many bookings

Availability and total calculation

A conflict exists when an existing confirmed booking starts before the requested return date and ends after the requested pickup date. Cancelled bookings are excluded. Both comparisons are strict, which permits adjacent reservations with no overlap. The date strings use YYYY-MM-DD, so validated values can be compared consistently.

```
existing.start_date < requested.end_date
AND existing.end_date > requested.start_date
```

```
days = (end_date - start_date) / 86,400,000
total = days * server_stored_daily_rate
```

The server begins an immediate transaction, repeats the overlap check, inserts the reservation, and commits. A conflict rolls back and returns HTTP 409. The stored rate and total preserve the original quote if the fleet rate is changed later. Integer NPR values avoid floating-point rounding in this model.

Implementation of the Application Layer

Table 4 API endpoints

Method and path	Access	Behavior
GET /api/cars	Public	Returns fleet and optional date availability.
POST /api/auth/register	Public	Validates identity fields and creates an account.
POST /api/auth/login	Public	Checks credentials and starts a session.
GET /api/auth/me	Signed in	Returns the current user identity.
POST /api/auth/logout	Signed in	Removes the session and clears its cookie.
GET /api/bookings	Signed in	Returns only the current user's reservations.
POST /api/bookings	Customer	Validates and transactionally reserves a vehicle.
PATCH /api/bookings/:id/cancel	Customer	Cancels a confirmed future reservation.
GET /api/health	Public	Returns an application health response.
GET /api/owner/bookings	Owner	Lists all customer booking orders.

Authentication and session handling

Registration requires a name, an email-shaped address, and an 8–128 character password. Email addresses are normalized to lowercase and must be unique. Passwords use scrypt with a random 16-byte salt, $N=131072$, $r=8$, $p=1$, and a 64-byte output. Authentication attempts are limited to 30 per 15-minute window per client address. This implementation uses synchronous hashing and is intended for modest demonstration traffic.

A successful authentication creates a random 32-byte token, stores its SHA-256 digest, and sets a seven-day session cookie. The raw token is not stored in the database. Protected handlers obtain the user identity from the session rather than accepting a user ID from the browser. Cancellation checks the booking ID and customer ID. Owner routes additionally require the owner role from the database; public registration cannot grant that role.

Frontend state and feedback

React state tracks the chosen trip, selected vehicle, authentication dialog, fleet filters, bookings, and messages. Date changes clear prior availability markers to avoid showing stale results. A guest can start a reservation and sign in without losing edited trip details. Busy states reduce repeated submissions; server errors remain visible when a reservation fails. Native dialog behavior and explicit focus restoration support keyboard use.

HTTP responses distinguish malformed input (400), missing authentication (401), failed verification or role checks (403), unavailable records (404), and reservation conflicts (409). Successful creation returns 201. The UI displays the corresponding message and keeps the user in the relevant workflow.

Project Structure and File Organization

Himalayan-Wheels/	
src/	
main.jsx	React components and API client
styles.css	Responsive site design
preview-api.js	Browser-only Pages adapter
OwnerDashboard.jsx	Customer order overview
shared/fleet.js	Shared demonstration fleet
server/	
app.js	Routes and booking rules
db.js	Schema and demo fleet
index.js	Startup and graceful shutdown
create-owner.js	Local owner setup command
tests/	
api.test.js	API integration scenario
docs/	
Himalayan-Wheels-Project-Report.pdf	
Project-Report.md	Editable report text
build_report.py	Report generation source
repository-qr.svg	Scannable repository link
screenshots/	Browser screenshots
.github/workflows/ci.yml	Automated verification
Dockerfile	Container starting configuration
.dockerignore	Excludes local build/data files
.gitignore	Excludes dependencies and data
index.html	Frontend entry point
vite.config.js	Build configuration and proxy
package.json	Scripts and dependency ranges
package-lock.json	Exact dependency resolutions
README.md	Setup, usage and API guide

Organization decisions

The client and server are separated by directory and communicate only through the API. The frontend uses reusable components for the logo, vehicle photos, modal, authentication, and booking form. The App component coordinates navigation and shared state. The database module initializes schema and fleet data; app.js defines API rules. OwnerDashboard.jsx renders orders, owners.js provisions owner accounts, and passwords.js shares password hashing.

Generated dependencies, frontend build output, local logs, and database files are excluded from Git. The source repository retains the dependency lockfile so another developer can reproduce the installation with npm ci. Tests create a disposable database in the operating system temporary directory, avoiding changes to the user's demonstration data.

A GitHub Actions workflow installs dependencies, runs the API tests, and builds the frontend on Node 24. The Dockerfile is supplied for a later hosting step. Its build and runtime behavior have not been executed as part of this native Windows verification, so it should be smoke-tested before deployment.

System Screenshots

The following figures were captured from the running application in a desktop browser. The visible fleet, prices, and accounts are demonstration data. Screenshots illustrate the actual implemented interface rather than a separate design mockup.

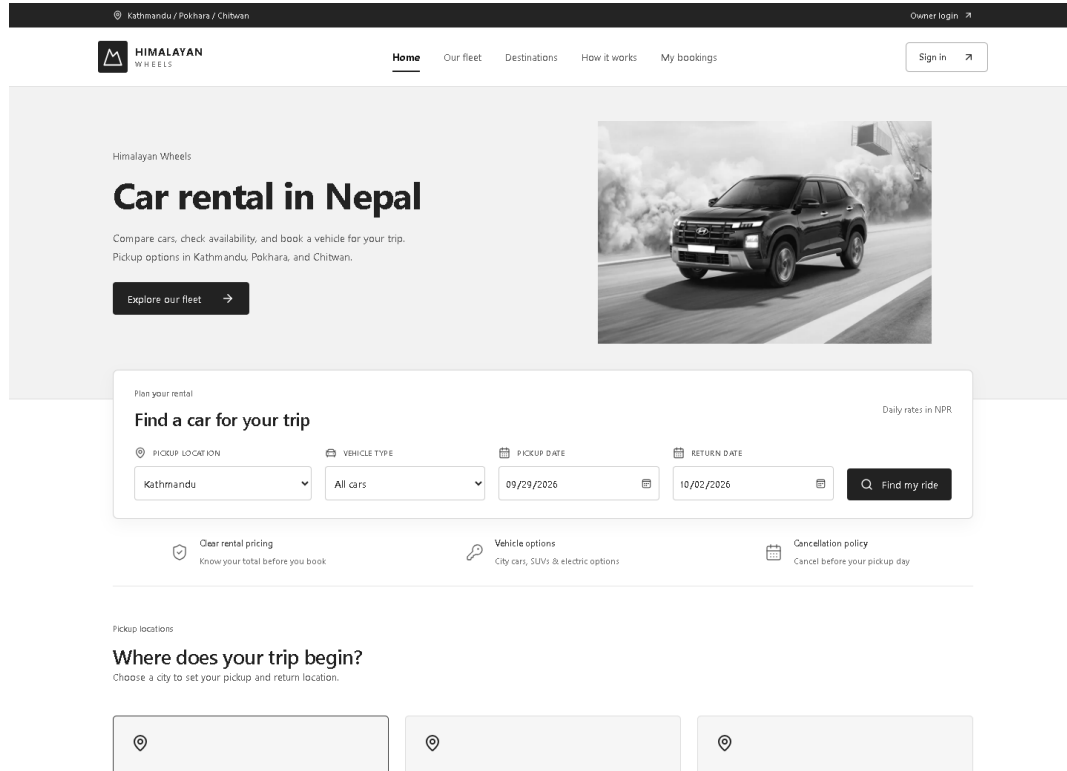


Figure 4 Homepage with rental search fields and navigation

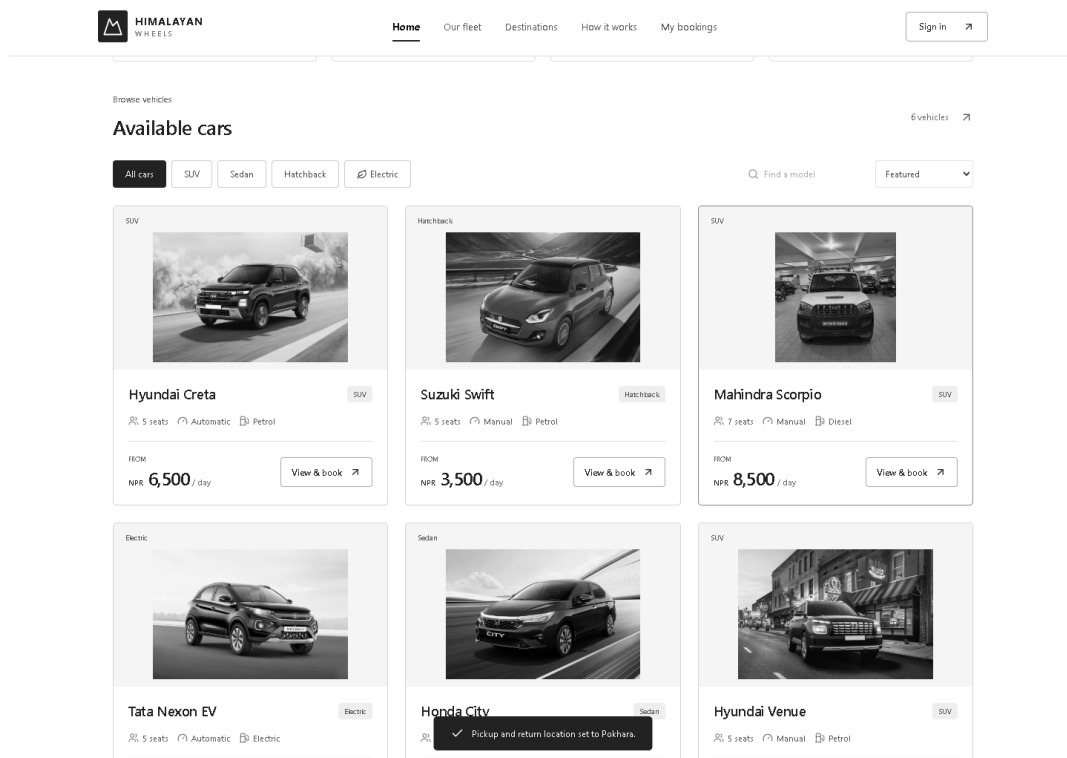
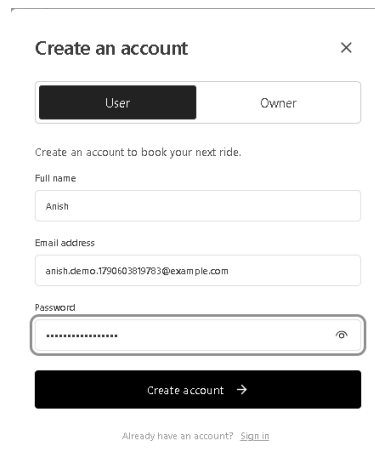


Figure 5 Vehicle cards with category controls, model search and pricing

System Screenshots of Registration and Reservation



Create an account

User Owner

Create an account to book your next ride.

Full name
Anish

Email address
anish.demo.1790403819783@example.com

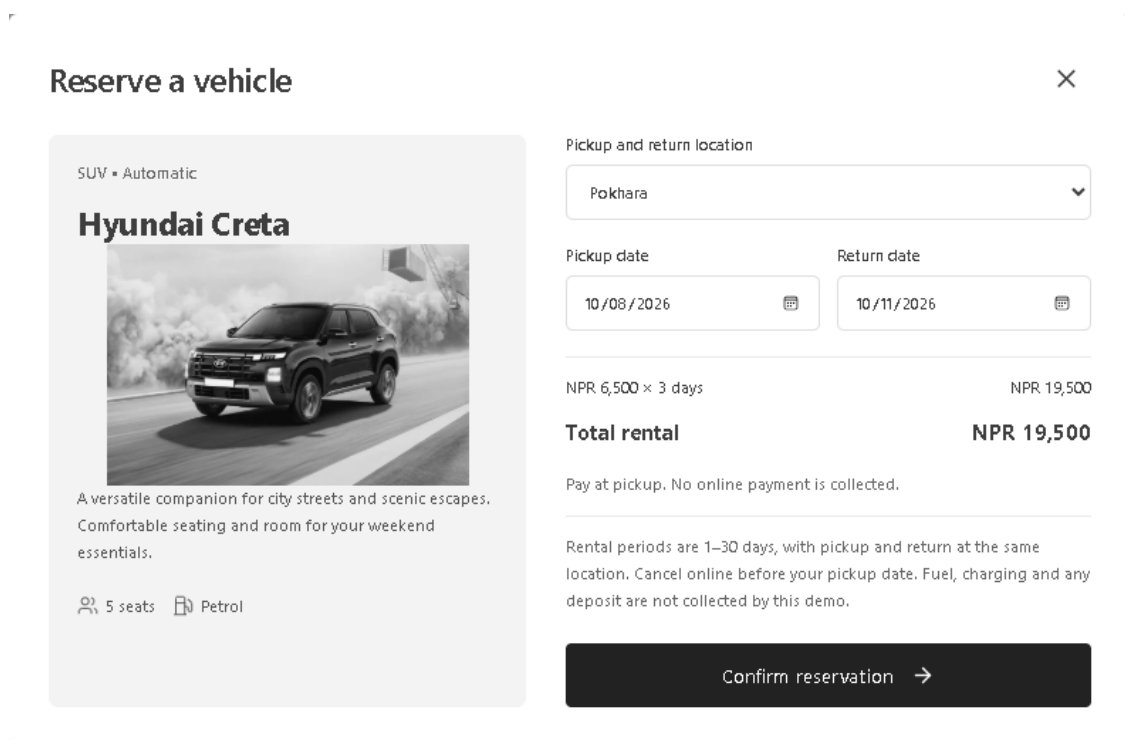
Password

Create account →

Already have an account? [Sign in](#)

Figure 6 Registration form with name, email and password inputs

A visitor can create an account within the booking journey. Password characters are obscured in the interface, and the API validates account details before starting a session.



Reserve a vehicle

SUV • Automatic

Hyundai Creta

A versatile companion for city streets and scenic escapes. Comfortable seating and room for your weekend essentials.

5 seats Petrol

Pickup and return location
Pokhara

Pickup date
10/08/2026

Return date
10/11/2026

NPR 6,500 × 3 days

NPR 19,500

Total rental **NPR 19,500**

Pay at pickup. No online payment is collected.

Rental periods are 1–30 days, with pickup and return at the same location. Cancel online before your pickup date. Fuel, charging and any deposit are not collected by this demo.

Confirm reservation →

Figure 7 Reservation dialog with dates, pickup city and total rental

The example shows a three-day Hyundai Creta booking at NPR 6,500 per day, producing NPR 19,500. The browser displays a preview, while the server recalculates the total from stored fleet data. The interface states that payment is at pickup and no online payment is collected.

System Screenshots of Bookings and Mobile Layout

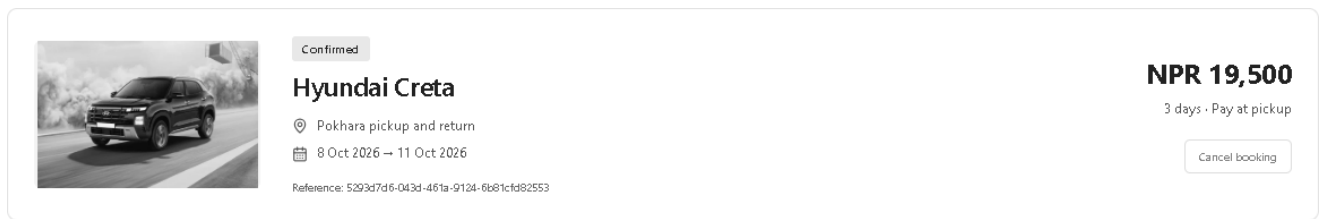


Figure 8 Private booking history and cancellation control

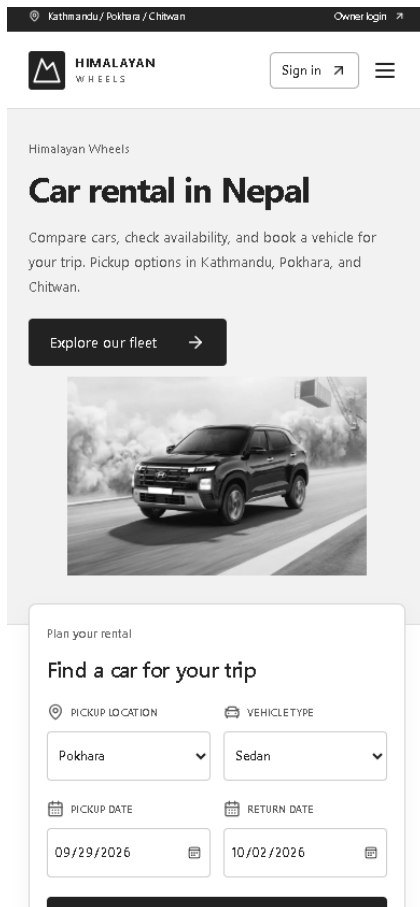


Figure 9 Mobile homepage

At a 390-pixel viewport, navigation moves into a menu, the vehicle photo is resized, and the search form becomes a compact stacked layout. Fleet cards become a single column.

Browser checks found no horizontal page overflow at this width. The same booking and account features remain accessible. This visual check does not replace a complete accessibility audit.

The booking history above shows the saved reference, rental dates, pickup city and total. Cancellation is offered only before the pickup date; cancelled reservations remain in history.

Challenges Faced

Maintaining consistent availability

A date search only provides a snapshot. Another request can reserve the car before confirmation. The solution is to recheck overlap inside the same transaction as the insert and return a conflict message when necessary. The test suite also submits competing requests and verifies that only one succeeds.

Preserving trip state across authentication

A guest may change the pickup location or dates in a booking dialog before signing in. Replacing that dialog with the authentication form initially risked resetting those edits. The final implementation copies the edited trip into parent state before opening authentication, then restores it when the reservation dialog returns. Browser verification confirmed that the selected city was retained.

Separating price previews from booking authority

The interface needs instant pricing feedback, but a submitted browser value cannot define the charge. Both layers compute the visible day count, while only the server reads the authoritative daily rate and persists the total. A test submits a false total of NPR 1 and verifies that the stored total remains NPR 19,500 for the three-day example.

Verification results

Table 5 Validation results

Area	Observed result
Authentication and ownership	Passed registration, login, logout, duplicate account and private-booking checks.
Dates and totals	Passed past-date, impossible-date, range-length and price-tampering checks.
Conflict handling	Passed overlapping, contained, surrounding, adjacent and competing reservation checks.
Cancellation	Passed owner-only cancellation, release of dates, repeat cancellation and pickup-day restriction.
Persistence	Passed booking and session retrieval after closing and reopening the server and database.
Browser workflow	Passed browsing, search, sort, account creation, booking, reload, cancellation and login/logout.
Mobile and build	No horizontal overflow at 390 px; no browser JavaScript errors; production build passed.

These checks establish the tested behavior, not production readiness. No load benchmark, penetration test, screen-reader audit, real payment integration, or commercial fleet validation was performed.

Conclusion

Himalayan Wheels implements the core customer journey of a car rental system: vehicle discovery, date selection, authentication, reservation, booking history, and cancellation. React delivers an interactive interface, Express enforces application rules, and SQLite retains the records required to connect users to vehicles over time.

The most significant result is the consistency between the visible workflow and the server-side rules. The price displayed to the customer is backed by a server calculation; availability is rechecked at confirmation; cancellation belongs to the reservation owner; and data survives a restart. The completed tests and browser checks provide evidence for these behaviors within the defined educational scope.

The application and report are suitable for local demonstration, source review, and further development. Publishing the repository supplies the code and documentation; running a public service requires a separate hosting and operational step. The project does not represent a real rental business, validated fleet inventory, or a payment-capable commercial platform.

Future Enhancements

- Staff operations: extend the read-only owner dashboard with audited fleet updates, maintenance blocks, pickup and return inspections, and controlled booking changes.
- Real inventory: model each vehicle's actual branch, relocation time, pickup and return timestamps, cleaning buffers, and location-dependent availability.
- Customer accounts: add verified email, password recovery, configurable cancellation terms, notification delivery, and downloadable booking documents.
- Payments: integrate a supported payment provider, verified callbacks, idempotency controls, deposits, refunds, and reconciliation before accepting real money.
- Database scaling: migrate to a coordinated server database and evaluate range exclusion constraints for conflict enforcement across multiple application instances.
- Accessibility and localization: add Nepali language support, screen-reader testing, contrast auditing, and broader device coverage.
- Operations: deploy behind HTTPS, establish persistent storage and backups, monitor errors, benchmark capacity, and perform a security review.

The suggested sequence is to strengthen real inventory and operational rules first, then improve identity and accessibility, and only then enable public transactions. Each expansion should introduce corresponding tests and revise the report's scope and assumptions.

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Source use

The cited documentation supports the technical design discussion. Package versions in the technology table come from the delivered dependency lockfile. Implementation descriptions and validation results refer to the delivered source and executed tests. Fleet data and rates are project examples. Sajilo Rental (<https://sajilorental.com/>) informed the search, destination, and category layout; business claims and customer reviews were not reused. Vehicle photographs are sourced from Meromoto (<https://meromoto.com/>), with individual source links in Image-Credits.md.

Appendix

Installation and first run

Install Node.js 24 or newer. Clone or download the repository, open a terminal in the project folder, and run the following commands. A network connection is needed to install packages. The built website uses local assets and a local database.

```
npm ci
npm run build
npm start
```

Open `http://localhost:3001`

Browse the fleet and select travel dates. Use Sign in and Create an account to register. Reserve a vehicle, inspect My bookings, and cancel a future booking if desired. To run the integration checks, execute `npm test`. For development, stop the current server and use `npm run dev`, then open `http://localhost:5173`.

Table 6 Runtime configuration

Variable	Default	Use
PORT	3001	Server port
HOST	127.0.0.1	Bind interface; containers usually use 0.0.0.0
DB_PATH	data/himalayan-wheels.sqlite	Persistent database file
NODE_ENV	Unset	production enables secure cookies; requires HTTPS
TRUST_PROXY	Unset	Use 1 only behind one trusted reverse proxy

Maintenance and troubleshooting

If the port is in use, stop the earlier server or select another PORT. If the frontend is absent, run `npm run build` before `npm start`. An experimental SQLite warning may appear on some Node 24 versions; inspect the actual server result rather than treating the warning as a test failure. If production cookies do not persist on plain HTTP, use HTTPS or run the local demonstration without `NODE_ENV=production`.

Stop the server before backing up the entire data folder. Do not commit the database, session records, or local secrets. A live deployment must preserve the database across releases. GitHub Pages cannot run this API; the repository is the publication destination for this delivery, and full-stack hosting is a separate future step. The Pages preview uses local browser storage and does not exercise the server authentication or shared reservation database.

Before academic submission, replace the cover-page fields for student name, roll number, institution, course, and supervisor. The editable Markdown report and the Python report source are included so the document can be customized.

Appendix: Owner and User Accounts

Local owner account setup

Run `npm run owner:create` in the project folder. Enter an owner name, a separate email, and a password of 12-128 characters. Password input is hidden and requires confirmation. If `DB_PATH` is customized, use the same value as the application server. Existing customer accounts are not promoted or overwritten.

Customers choose User in the sign-in form and create an account to reserve a car. Owners choose Owner login and enter the credentials created above. Public registration always stores the customer role. Existing databases are migrated without removing users or reservations.

The screenshot displays the 'Owner dashboard' for 'Himalayan Wheels'. The top navigation bar includes links for Home, Our fleet, Destinations, How it works, and Owner orders (which is the active page). A 'Refresh orders' button is located in the top right corner. The main content area is titled 'Booking orders' and includes a sub-header 'View customer reservations and their current status.' Below this, a summary table shows: Total orders: 1, Confirmed: 1, Cancelled: 0, and Confirmed booking value: NPR 19,500 (Booked value, not payments received). A search bar and a 'Booking status' dropdown (set to 'All statuses') are provided. A table lists 1 of 1 orders, showing a booking for a 'Hyundai Creta' from '29 Sept 2026 to 2 Oct 2026' (3 days) in 'Kathmandu', with a total of '19,500 NPR 6,500 / day'. The status is 'Confirmed'. The footer contains links for 'EXPLORE' (Our fleet, How it works, Owner orders), 'PICKUP CITIES' (Kathmandu, Pokhara, Chitwan), and 'THE PROJECT' (View on GitHub, Academic demonstration, No real rental or payment is issued.). A 'Welcome, Demo.' notification is visible in the bottom center.

Kathmandu / Pokhara / Chitwan

Owner dashboard

HIMALAYAN WHEELS

Home Our fleet Destinations How it works **Owner orders** Hi, Demo

Owner account

Booking orders

View customer reservations and their current status.

Refresh orders

Total orders	Confirmed	Cancelled	Confirmed booking value
1	1	0	NPR 19,500
			Booked value, not payments received

Search orders

Customer, email, car or reference

Booking status

All statuses

1 of 1 orders

Customer bookings - amounts in NPR

Customer	Vehicle / reference	Rental dates	Pickup / return	Total	Status
Demo customer customer@example.com	Hyundai Creta 75ebd9b3-a696-4133-a8f2-4bfac6dd3d05	29 Sept 2026 to 2 Oct 2026 3 days	Kathmandu	19,500 NPR 6,500 / day	Confirmed

HIMALAYAN WHEELS

Car rental in Nepal.
Kathmandu, Pokhara and Chitwan.

EXPLORE

Our fleet

How it works

Owner orders

PICKUP CITIES

Kathmandu

Pokhara

Chitwan

THE PROJECT

View on GitHub

Academic demonstration.

No real rental or payment is issued.

Welcome, Demo.

Figure 10 Owner dashboard showing a customer booking order

The owner dashboard includes customer names and emails, vehicles, dates, location, totals, and status. Owners can search, filter by status, and refresh orders. Confirmed booking value is not payment revenue. Server checks reject unauthenticated access and customer access to the owner endpoint.

GitHub demo and verification

The Pages demo offers User and Owner choices without passwords. Create a booking as the demo traveller, sign out, and continue as the demo owner in the same browser. These are browser-only sample orders. Tests cover real owner login, customer isolation, role tampering, database migration, saved orders after restart, cancellation visibility, and responsive browser interaction.